

Strong Convergence in L^2 Using Compactness

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A key step in establishing global regularity for the Navier–Stokes equations is demonstrating strong convergence in L^2 . Compactness arguments play a central role in achieving this result, bypassing classical Sobolev embedding limitations through the integration of spectral decay and energy boundedness.

1.1 Compactness Framework

The compactness framework leverages the generalized solution space H :

$$\|u\|_H = \|u\|_{L^2} + \|\nabla u\|_{L^2} + \sup_{k>k_0} \frac{E(k)}{k^\beta}, \quad \beta > 0. \quad (1)$$

This norm ensures control over low-frequency, intermediate, and high-frequency energy contributions.

Properties of the H -Norm:

- **Low-Frequency Control:** $\|u\|_{L^2}$ bounds the total kinetic energy.
- **Intermediate-Frequency Control:** $\|\nabla u\|_{L^2}$ ensures finite enstrophy.
- **High-Frequency Suppression:** $\sup_{k>k_0} \frac{E(k)}{k^\beta}$ enforces spectral decay, aligning with the Scale-Barrier Principle.

1.2 Compactness in the Space H

The compactness result is formalized as follows:

Theorem 1.1 (Compactness in L^2). *Let $\{u_n\} \subset H$ be a sequence of approximate solutions to the Navier–Stokes equations, satisfying:*

1. $\sup_n \|u_n\|_H < \infty$.
2. $\partial_t u_n$ is uniformly bounded in $L^2(0, T; H^{-1})$.

Then, $\{u_n\}$ has a subsequence that converges strongly in $L^2(0, T; L^2)$.

Proof. The proof follows the adaptation of the Aubin–Lions lemma using spectral decay:

1. **Uniform Boundedness:** The assumption $\sup_n \|u_n\|_H < \infty$ implies uniform control over $\|u_n\|_{L^2}$, $\|\nabla u_n\|_{L^2}$, and $\sup_{k>k_0} \frac{E_n(k)}{k^\beta}$.

2. ****Time Derivative Control:**** Boundedness of $\partial_t u_n$ in $L^2(0, T; H^{-1})$ ensures weak compactness in time.
3. ****Spectral Decay:**** The Scale-Barrier Principle guarantees that:

$$E(k) \leq Ck^{-\gamma}, \quad \gamma > \beta + 2. \quad (2)$$

This suppresses high-frequency energy contributions.

4. ****Strong Convergence:**** Combining weak compactness with spectral control, we conclude strong L^2 convergence:

$$\|u_n - u\|_{L^2(0, T; L^2)} \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (3)$$

□

1.3 Role of Compactness in Global Regularity

Compactness ensures:

- ****Existence of Smooth Limits:**** Approximate solutions converge to smooth limits, supporting the passage to the weak formulation.
- ****Control of High-Frequency Energy:**** The spectral decay condition suppresses turbulence-like cascades, preventing blowup.
- ****Integration with Other Principles:**** Compactness complements energy boundedness ($Q(t)$) and curvature-based gradient control.

1.4 Numerical Validation Framework

Simulation Setup: To validate compactness numerically:

- ****Initial Data:**** Choose $u_0 \in H^1$ with finite $\|u_0\|_{H^1}$.
- ****Forcing Term:**** Use $f \in L^2$ to sustain energy input.
- ****Discretization:**** Apply spectral methods with resolution satisfying $\Delta x \ll \eta$, where η is the Kolmogorov scale.

Metrics: Track:

1. ****Strong Convergence:**** Compute $\|u_n - u\|_{L^2}$ over time and verify decay.
2. ****Spectral Energy Decay:**** Validate $E(k) \sim k^{-\beta}$ for high wave numbers.
3. ****Compactness Indicators:**** Ensure uniform bounds on $\sup_{k > k_0} \frac{E(k)}{k^\beta}$.

Conclusion

Compactness in L^2 ensures strong convergence of approximate solutions, addressing key obstacles to global regularity for the Navier–Stokes equations. By integrating spectral decay, energy boundedness, and gradient control, it forms a critical link in the unified proof.

References

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